

1. (a) $A, B \in \mathcal{C} \Rightarrow A \cup B \in \mathcal{C}; A \in \mathcal{C} \Rightarrow A^C \in \mathcal{C}; \Omega \in \mathcal{C}$.
 (b) Only if part is proved in class. For the if part, $A \in \mathcal{C} \Rightarrow A^C = \Omega \setminus A \in \mathcal{C}$. Also, if $A, B \in \mathcal{C}$, then $A \cap B = A \cap [(B^C)^C] = A \setminus B^C \in \mathcal{C}$, since $B^C \in \mathcal{C}$. Hence, by DeMorgan's we have that $\mathcal{C}$ is an algebra.
 (c) Checking $\mathcal{A}_i$ are algebras is easy. Note $\{a, b, c\} = \{a, b\} \cup \{a, c\}$ is not in the union.
 (d) $\Omega \in \mathcal{A}$, since $\Omega \in \mathcal{A}_n$ for all $n \geq 1$. If $A \in \mathcal{A}$, then $A \in \mathcal{A}_{n_0}$ for some n_0 . In that case $A^c \in \mathcal{A}_{n_0} \subset \mathcal{A}$. If $A, B \in \mathcal{A}$, then there exists $n_1, n_2 \geq 1$, such that $A \in \mathcal{A}_{n_1}, B \in \mathcal{A}_{n_2}$. Then for $n = \max(n_1, n_2) \geq 1$, we have $A, B \in \mathcal{A}_n \subset \mathcal{A}$.
2. (a) Using monotonicity, $P(A \setminus B) \leq P(A \Delta B) = 0$. So, By finite additivity $P(A) = P(A \setminus B) + P(A \cap B) = P(A \cap B)$. Similarly $P(B) = P(A \cap B)$.
 (b) Taking $A_i = B_i^c$, we have $P(A_i) = 0$. Using countable subadditivity, $P(\cup A_i) \leq \sum P(A_i) = 0$. So by deMorgan's law, $P(\cap B_i) = 1 - P(\cup A_i) = 1$.
 (c) $f(x) = \sin(x) + \cos(x)$ is continuous function, hence measurable. So $Y = f(X)$ is measurable. Hence $A = Y^{-1}(\{1\}) \in \mathcal{F}$.
3. (a) Let $\mathcal{C} = \{(a, b] : -\infty \leq a < b < \infty\} \cup \{(a, \infty), a \in \mathbb{R}\}$. It is a semi-algebra. Define $\mu_F(a, b] = F(b+) - F(a+) = F(b) - F(a)$; $\mu_F(a, \infty] = F(\infty-) - F(a+) = 1 - F(a)$. Then this defines a measure on semi-algebra $\mathcal{C}$. By Caratheodory Extension Theorem, there exists an extension of μ_F , called μ_F^* on $(\Omega, \mathcal{M}_{\mu_F^*, \mu_F^*})$, where $\mathcal{M}_{\mu_F^*}$ = set of all μ_F^* -measurable sets on $\Omega = \{A \in \mathcal{P}(\Omega) : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^C), \forall E \in \Omega\}$. $\mathcal{M}_{\mu_F^*}$ is a σ -algebra containing $\mathcal{C} \Rightarrow \mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subset \mathcal{M}_{\mu_F^*}$. μ_F^* restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an extension of μ_F .
 (b) This is unique by the uniqueness theorem, since $\mu_F^*(\mathbb{R}) = 1$ (finite measure, hence any other measure that takes same value on $\mathcal{C}$ will take same values on $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$).
 (c) $\mu_F^*(A) = \mu_F^*(2) - \mu_F^*(0.5) = 1 - 0.5^3 = 0.875$.